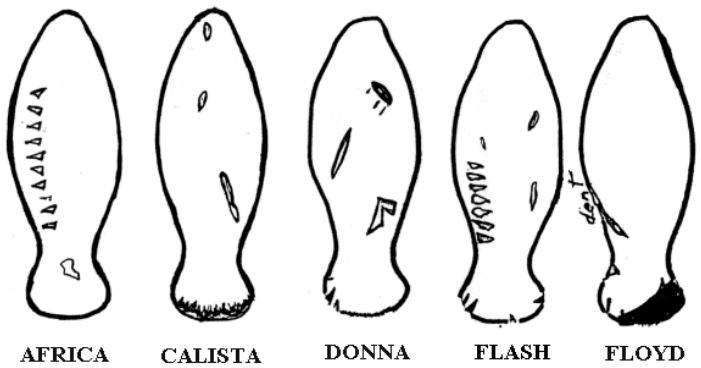
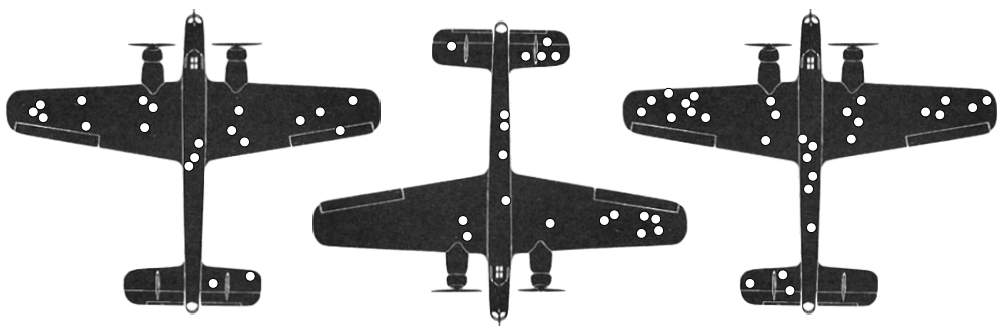
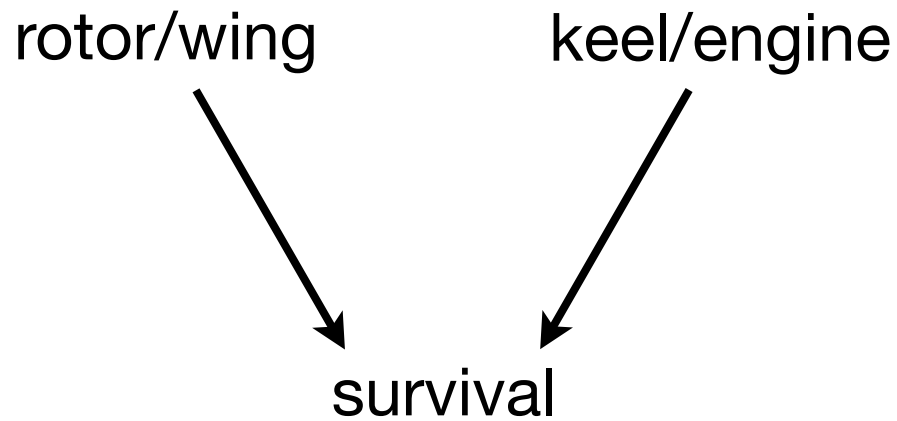






Observe only:
undamaged
rotor/wing damaged

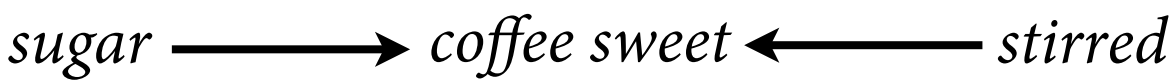
Figure 8.1

Interaction effects

- *Interactions*: Influence of predictor conditional on other predictor(s)
  - Influence of *sugar* in *coffee* depends on *stirring*
  - Influence of *gene* on *phenotype* depends on *environment*
  - Influence of *skin color* on *cancer* depends on *latitude*
- Generalized linear models (GLMs): All predictors interact to some degree
- Multilevel models: Massive interaction engines

Interaction effects in DAGs

- In DAG, interaction doesn't look special:



- This just means:

$$coffee\ sweet = f(sugar, stirred)$$

- We have to figure out the function f.

Interpreting interactions

- Is hard
  - Add interaction => other parameters change meaning
  - Influence of predictor depends upon multiple parameters and their covariation

Additionally, you need a sample that's 16x larger to measure the same size of an effect as for a beta parameter!

R code 8.14

```
precis( m8.5 , depth=2 )
```

```
        mean   sd  5.5% 94.5%
a[1]    0.89 0.02  0.86  0.91
a[2]    1.05 0.01  1.03  1.07
b[1]    0.13 0.07  0.01  0.25
b[2]   -0.14 0.05 -0.23 -0.06
sigma   0.11 0.01  0.10  0.12
```

Tulip model – interaction

R code 8.24

```
m8.7 <- quap(
    alist(
        blooms_std ~ dnorm( mu , sigma ) ,
        mu <- a + bw*water_cent + bs*shade_cent + bws*water_cent*shade_cent ,
        a ~ dnorm( 0.5 , 0.25 ) ,
        bw ~ dnorm( 0 , 0.25 ) ,
        bs ~ dnorm( 0 , 0.25 ) ,
        bws ~ dnorm( 0 , 0.25 ) ,
        sigma ~ dexp( 1 )
    ) ,
    data=d )
```

Interpreting parameters very hard! Plot.

Posterior predictions

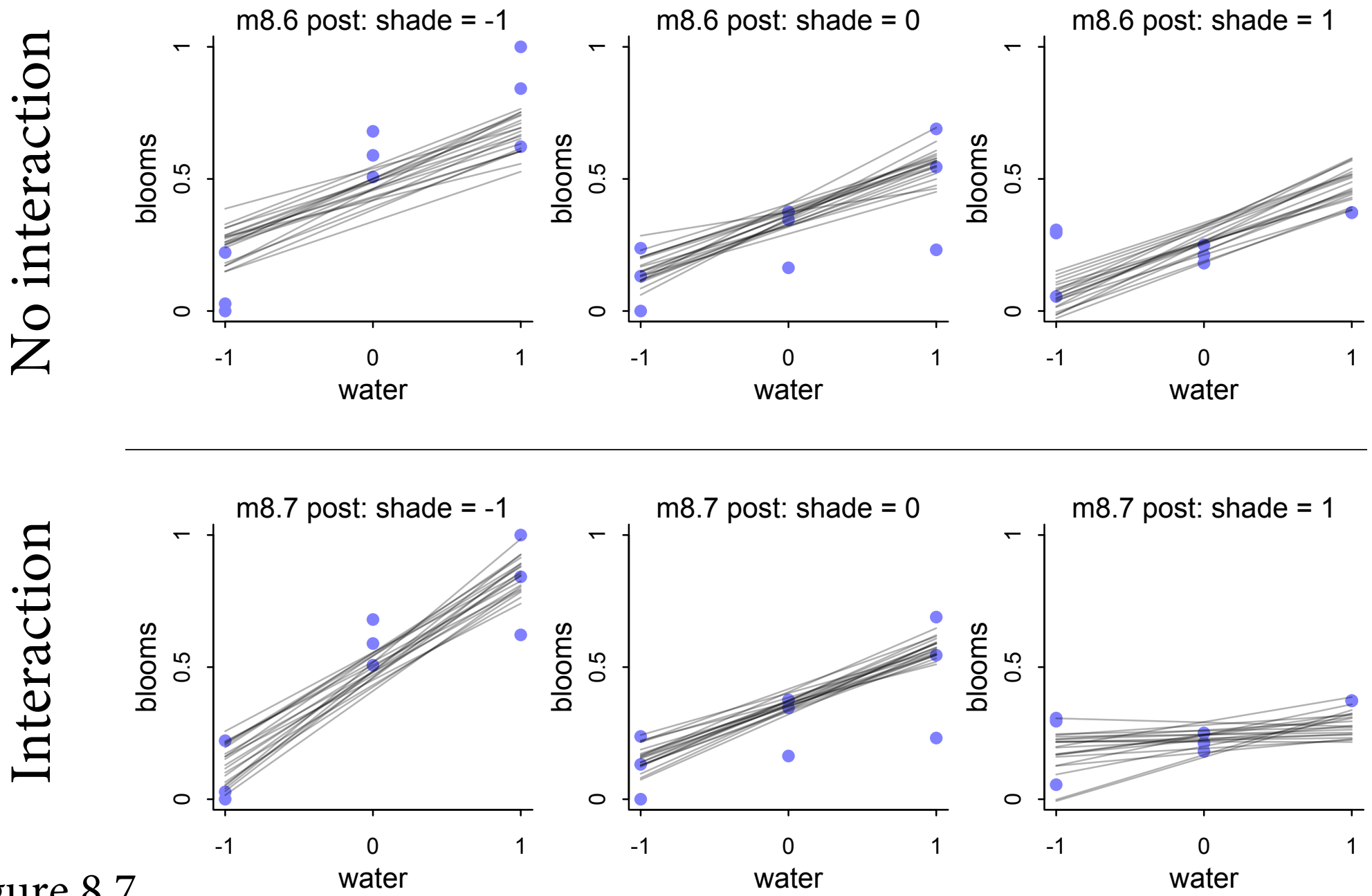
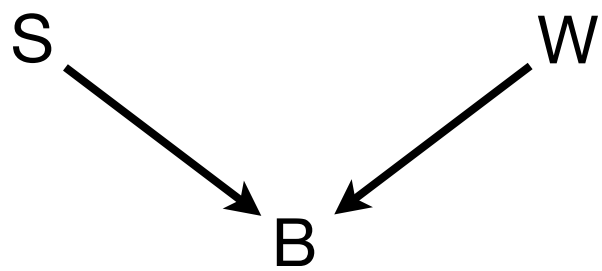


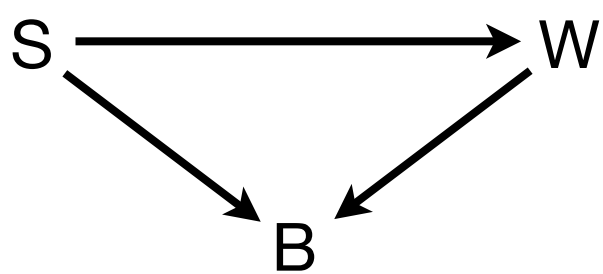
Figure 8.7

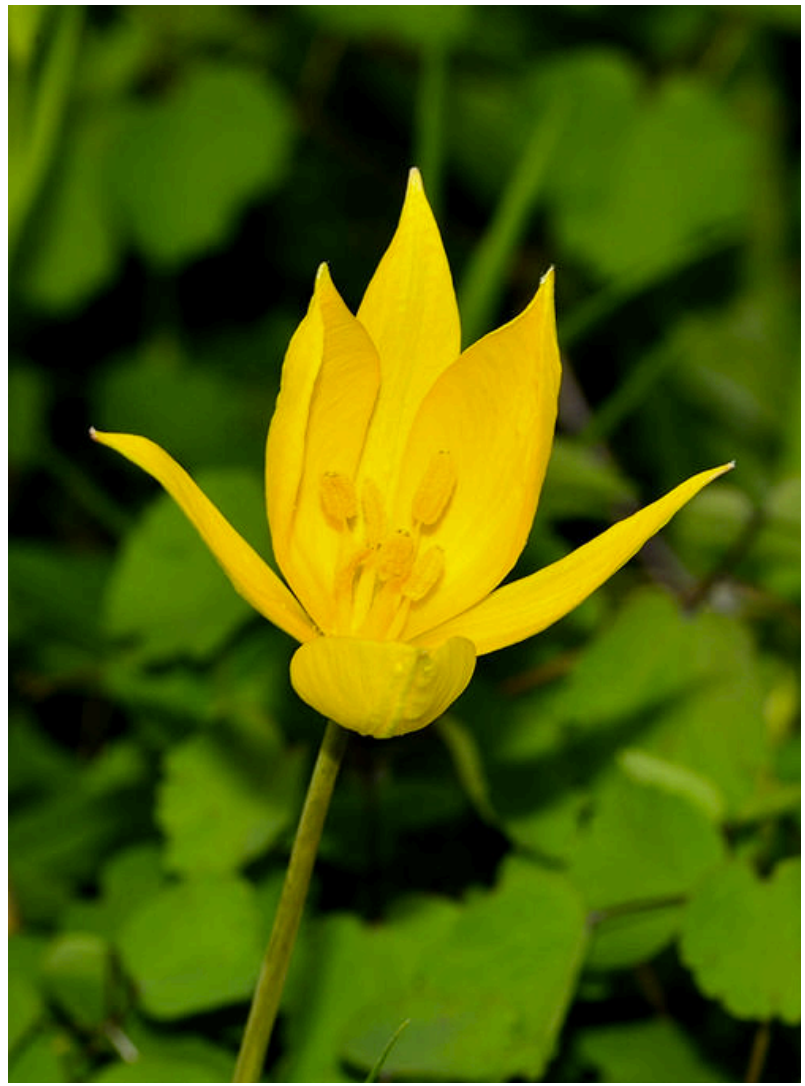
Causal thinking

- Tulip experiment:



- Tulip reality:



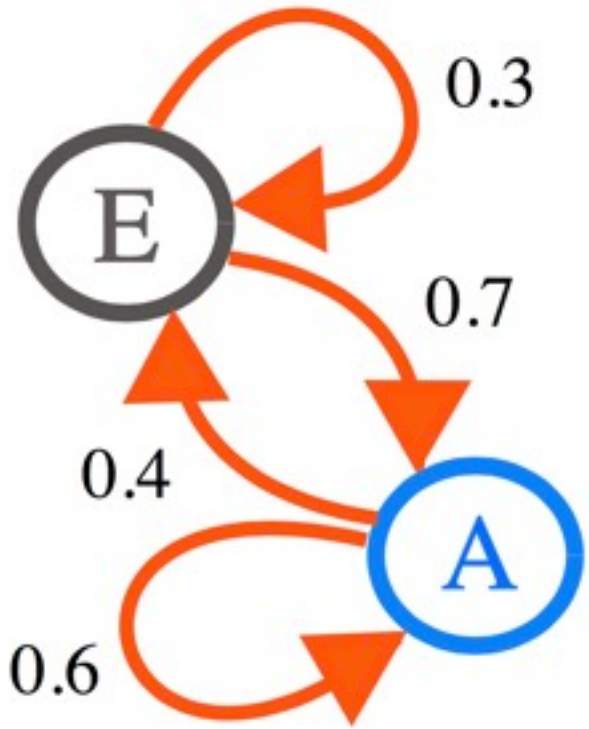


Interactions not always linear

- Suppose all tulip data collected under “cool” temperatures
- Under “hot” temperature, tulips do not bloom
- Interaction, but not a linear one
  - blooms goes to zero at threshold

Markov chain Monte Carlo

- Markov chain Monte Carlo (MCMC)
  - Understand the approach
  - Meet different algorithms
- Interfaces to MCMC: Stan & `ulam`
- How to sample responsibly
- How to recognize and fix problems



Why MCMC?

- Sometimes can't write an integrated posterior
- Even when can, often cannot use it
- Many problems are like this: Multilevel models, networks, phylogenies, spatial models
- Optimization not a good strategy in high dimensions — must have full distribution
- **MCMC is not fancy. It is old and essential.**

Hamiltonian Monte Carlo

- Problem with Gibbs sampling (GS)
  - High dimension spaces are *concentrated*
  - GS gets stuck, degenerates towards random walk
  - Inefficient because re-explores
- Hamiltonian dynamics to the rescue
  - represent parameter state as particle
  - flick it around frictionless log-posterior
  - record positions
  - no more "guess and check"
  - all proposals are good proposals

William Rowan Hamilton (1805–1865) Commemorated on Irish Euro coin

THE #1 BAYESIAN EXCUSE FOR LEGITIMATELY SLACKING OFF:

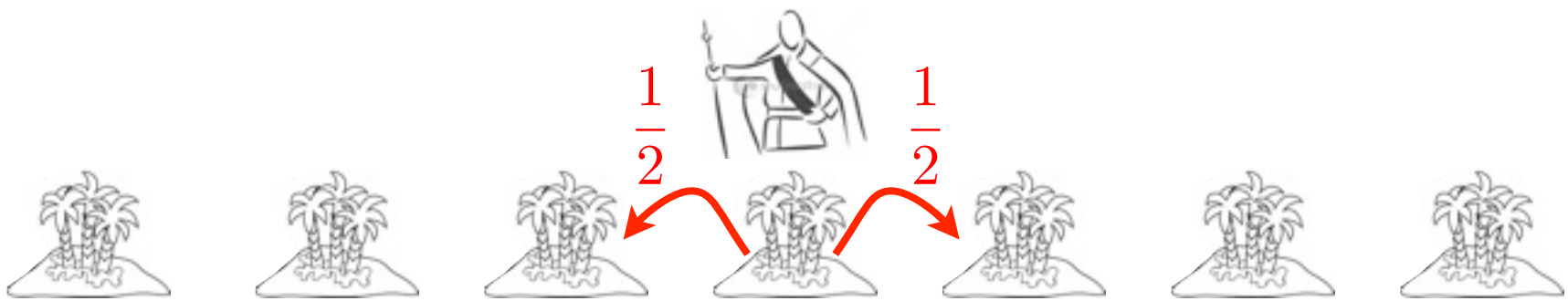
"MCMC RUNNING."

King Markov

The Metropolis Archipelago

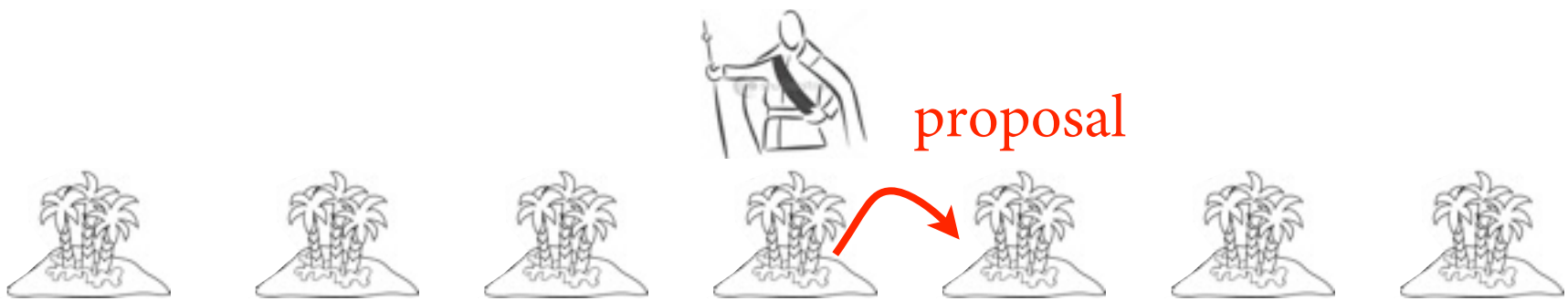
Contract: King Markov must visit each island in proportion to its population size.

Here's how he does it...



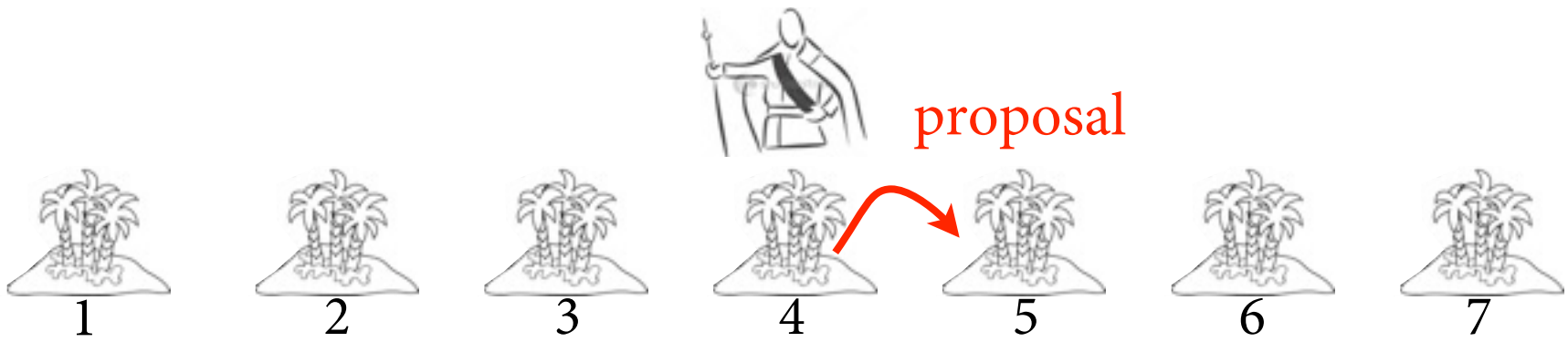
(1) Flip a coin to choose island on left or right.
Call it the “proposal” island.

(1) Flip a coin to choose island on left or right.
Call it the "proposal" island.



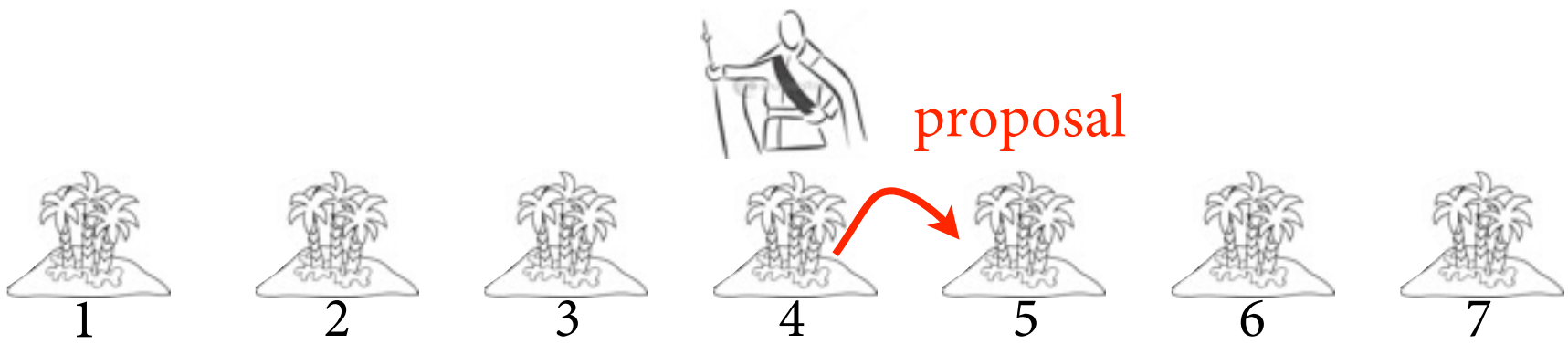
(2) Find population of proposal island.

(1) Flip a coin to choose island on left or right.
Call it the "proposal" island.



(2) Find population of proposal island.

(1) Flip a coin to choose island on left or right.
Call it the "proposal" island.

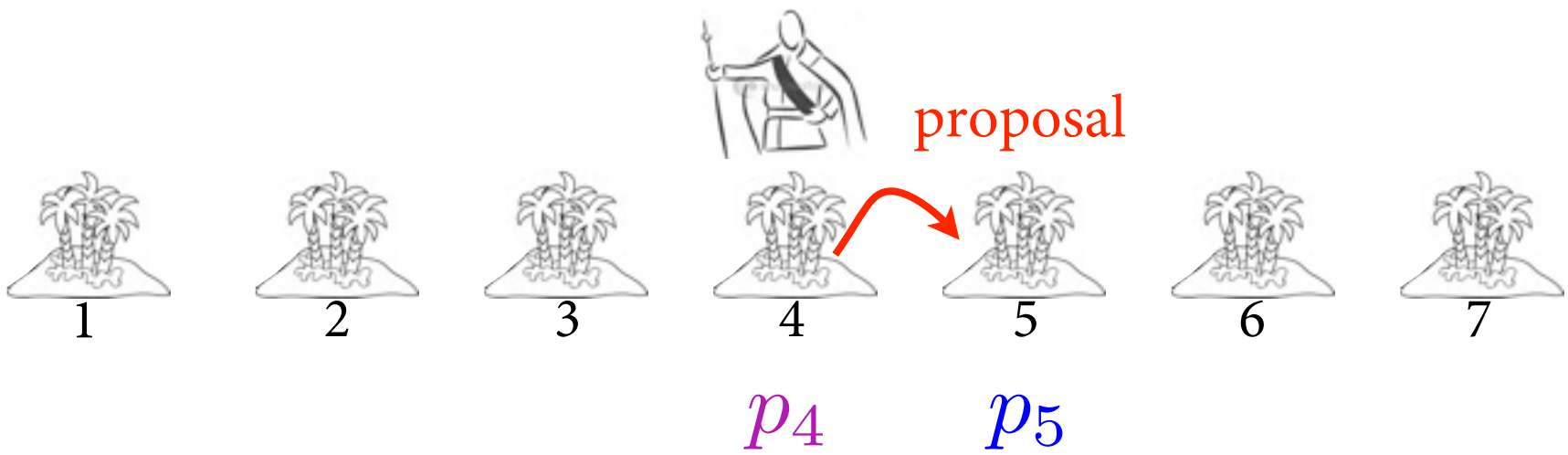


p_5

(2) Find population of proposal island.

(1) Flip a coin to choose island on left or right.
Call it the "proposal" island.

(2) Find population of proposal island.

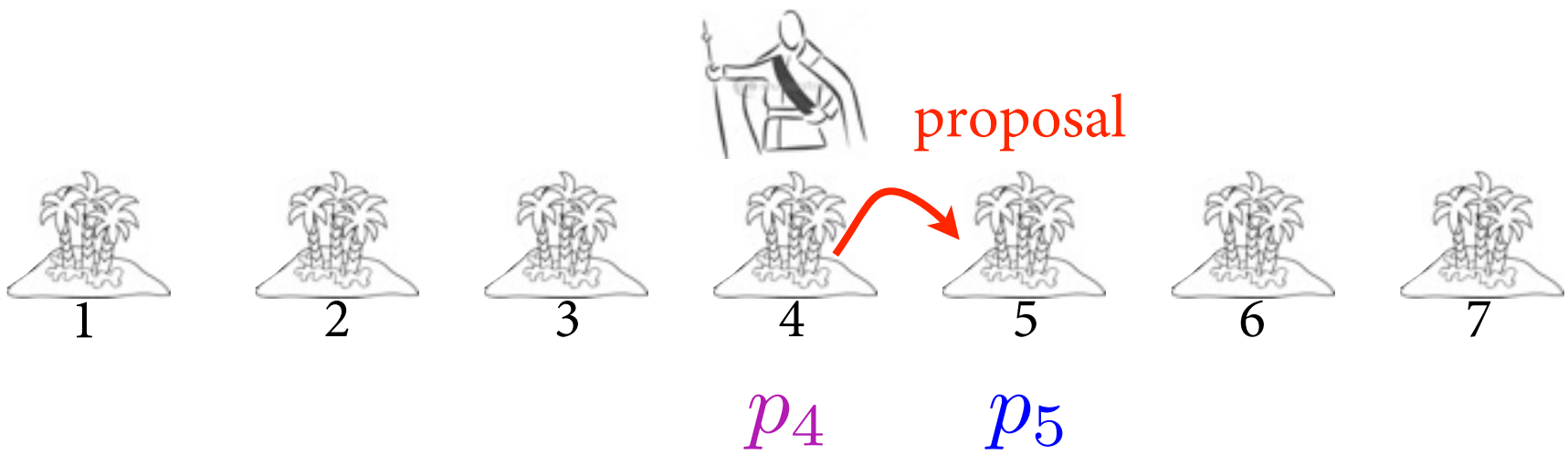


(3) Find population of current island.

(1) Flip a coin to choose island on left or right.
Call it the "proposal" island.

(2) Find population of proposal island.

(3) Find population of current island.



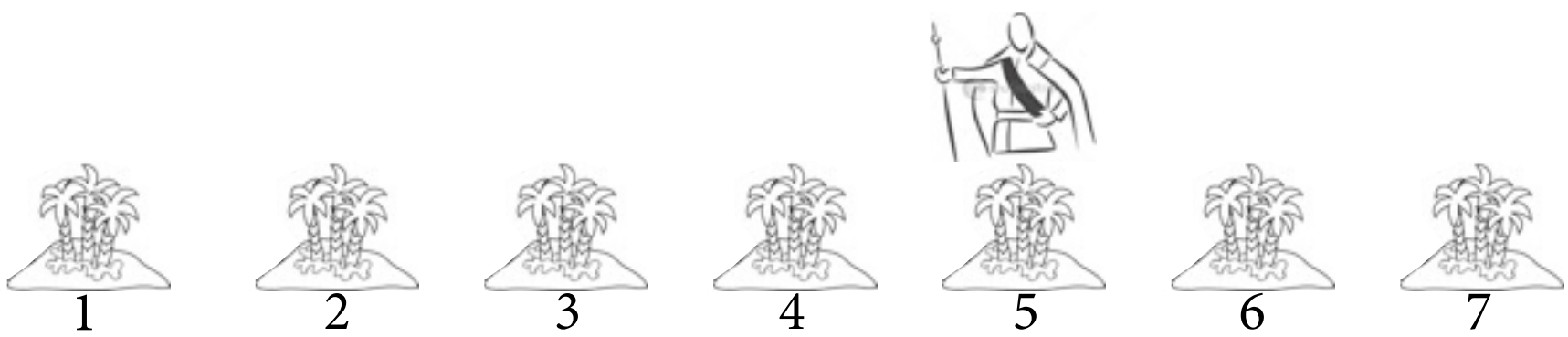
(4) Move to proposal, with probability = $\frac{p_5}{p_4}$

(1) Flip a coin to choose island on left or right.
Call it the "proposal" island.

(2) Find population of proposal island.

(3) Find population of current island.

(4) Move to proposal, with probability = $\frac{p_5}{p_4}$



(5) Repeat from (1)

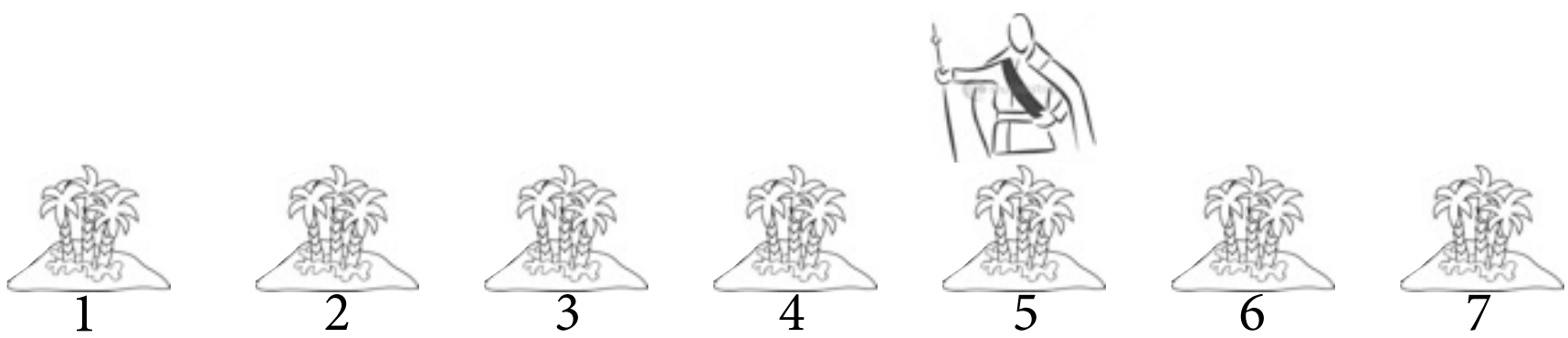
(1) Flip a coin to choose island on left or right.
Call it the "proposal" island.

(2) Find population of proposal island.

(3) Find population of current island.

(4) Move to proposal, with probability = $\frac{p_5}{p_4}$

(5) Repeat from (1)



This procedure ensures visiting each island in proportion to its population, *in the long run.*

https://chi-feng.github.io/mcmc-demo/

Hamiltonian Flows

- What happens when you use ulam?
  - Translates formula into raw Stan model code
  - Stan then builds a custom NUTS sampler
  - Sampler runs
  - Samples fed back to R

```
m9.1 <- ulam(
    alist(
        log_gdp_std ~ dnorm( mu , sigma ) ,
        mu <- a[cid] + b[cid]*( rugged_std - 0.215 ) ,
        a[cid] ~ dnorm( 1 , 0.1 ) ,
        b[cid] ~ dnorm( 0 , 0.3 ) ,
        sigma ~ dexp( 1 )
    ) ,
    data=dat_slim , chains=4 , cores=4 , iter=1000 )
```

R code 9.14

Hamiltonian Flows

R code 9.16

```
precis( m9.1 , 2 )
```

```
       mean   sd  5.5% 94.5% n_eff Rhat
sigma  0.11 0.01  0.10  0.12  2641    1
b[1]   0.13 0.07  0.02  0.26  2484    1
b[2]  -0.14 0.05 -0.23 -0.06  2546    1
a[1]   0.89 0.02  0.86  0.91  3331    1
a[2]   1.05 0.01  1.03  1.07  3243    1
```

- n_eff: number of effective samples
  - can be larger than actual samples!
- Rhat: Convergence diagnostic — “1” is good

```
pairs( m9.1 )
```

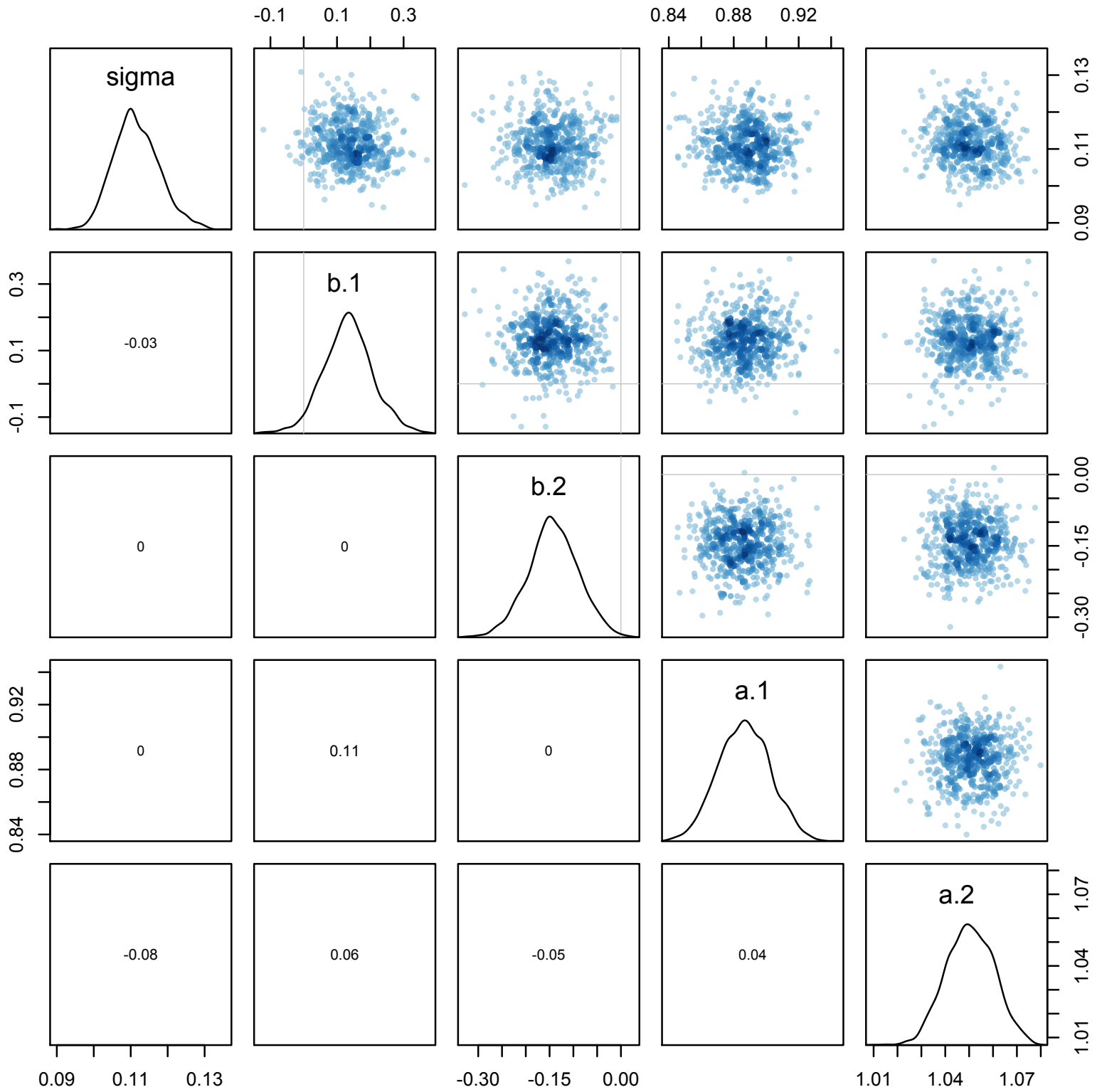
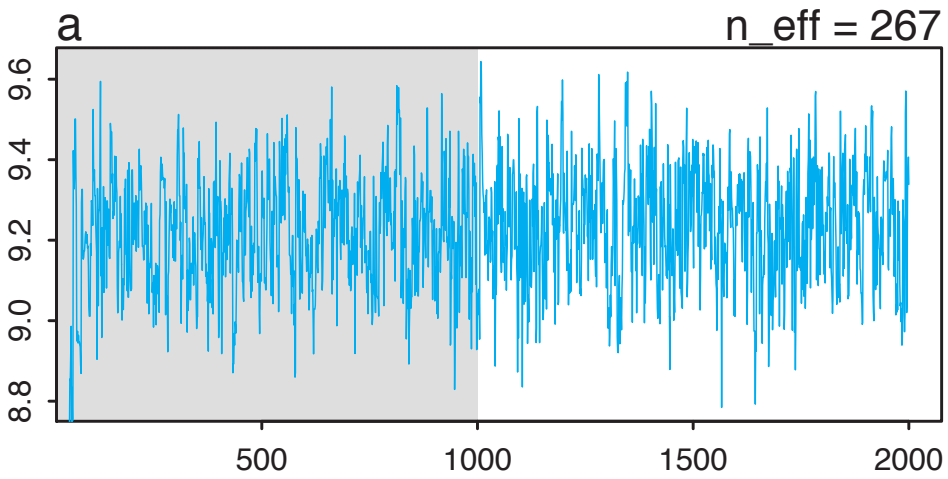
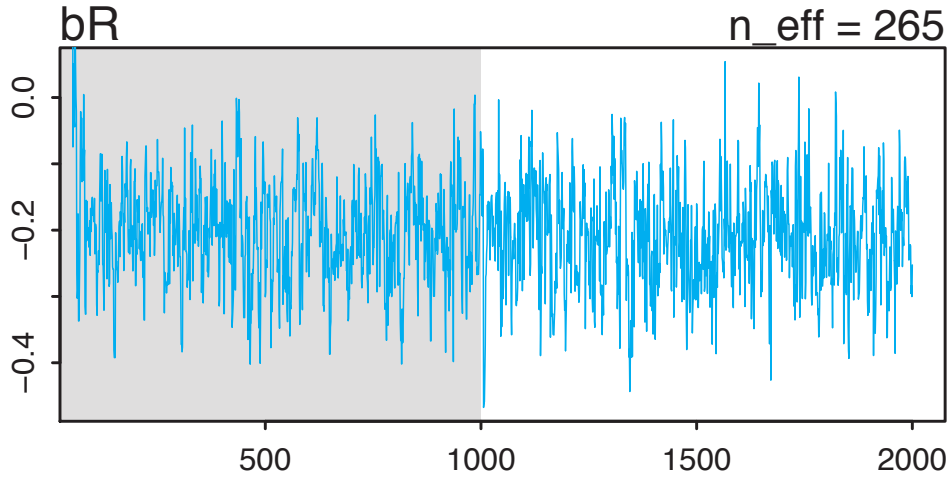
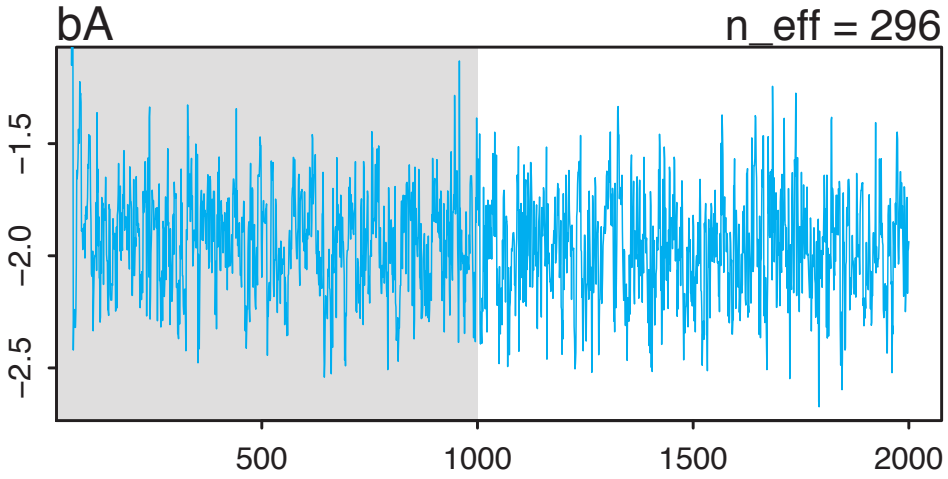
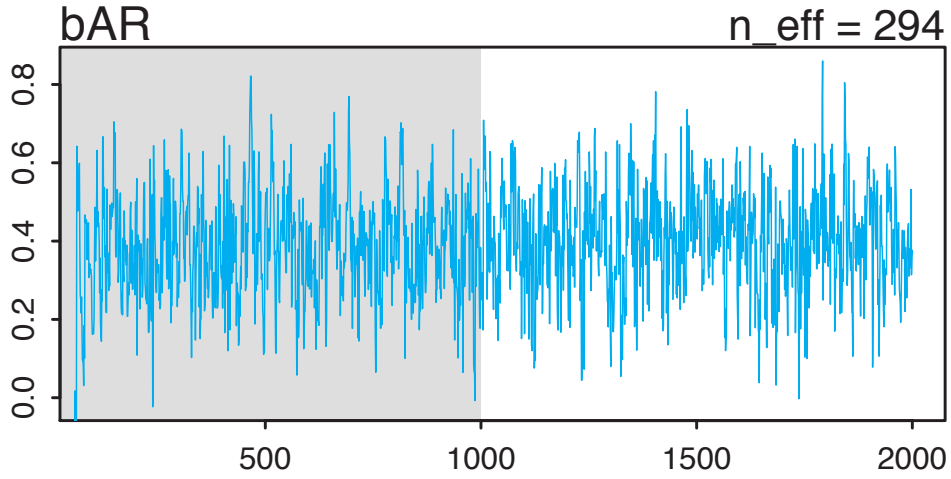


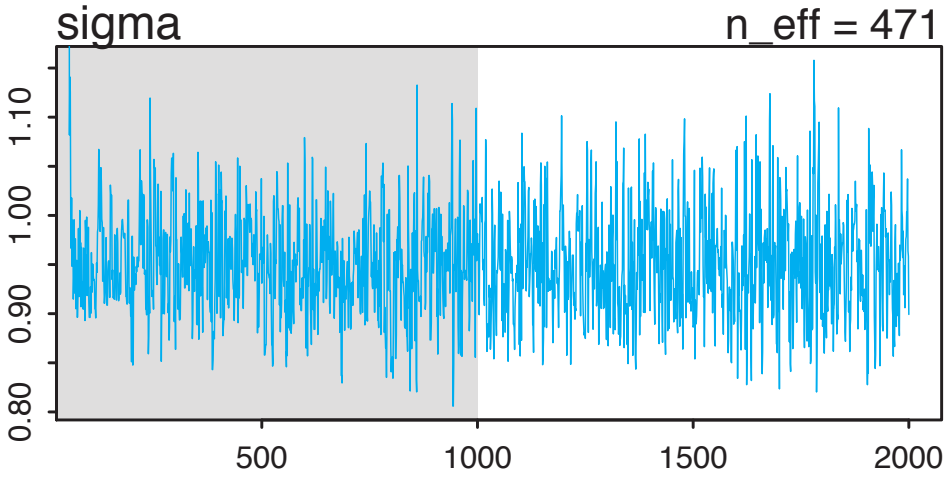
Figure 9.7











"Hairy caterpillar ocular inspection test"

Convergence diagnostics

- n_eff: "effective" number of samples
  - n_eff/n < 0.1, be alarmed
- R-hat
  - R-hat: crudely, ratio of variance between chains to variance within chains
  - Should approach 1
- Both may mislead

R code 9.16

```
precis( m9.1 , 2 )
```

```
       mean   sd  5.5% 94.5% n_eff Rhat
sigma  0.11 0.01  0.10  0.12  2641    1
b[1]   0.13 0.07  0.02  0.26  2484    1
b[2]  -0.14 0.05 -0.23 -0.06  2546    1
a[1]   0.89 0.02  0.86  0.91  3331    1
a[2]   1.05 0.01  1.03  1.07  3243    1
```


May the effective sampe size be great!
Let my Rhat converge to 1.0
I pray my chains have mixed well

The Folk Theorem of Statistical Computing

"When you have computational problems, often there's a problem with your model."

–Andrew Gelman